BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: KT-XI-23-I

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** Code B

Note: Q.1 Attempt all question of Section A by filling the correct option i.e. A, B, C or D on the OMR RESPONSE SHEET. No marks will be awarded for cutting, erasing or over writing.

Qi. As compared to calculated value, the experimental bond length of a polar diatomic molecule is ....

(A) greater
(B) shorter
(C) negligible
(D) zero

Qii. Bond order gives information about all of the following except ....

(A) stability
(B) bond energy
(C) bond length
(D) paramagnetism

Qiii. N_2 and CO have the same rates of diffusion because both are ...

(A) gases
(B) less reactive
(C) more reactive
(D) of same masses

Qiv. Spreading of gas in air is due to .....

(A) lower density
(B) diffusion
(C) effusion
(D) attraction with air

Qv. A catalyst increases the rate of a chemical reaction by lowering the .....

(A) activation energy
(B) internal energy
(C) Gibb's free energy
(D) kinetic energy

Qvi. A crystalline solid can contain maximum of only one....

(A) Axis of symmetry
(B) plane of symmetry
(C) center of symmetry
(D) both A&B

Qvii. Pin – wheel experiment showed that cathode rays have ....

(A) particle nature
(B) wave nature
(C) positive charge
(D) negative charge

Qviii. Water and ethyl alcohol are completely miscible into each other due to

(A) polar nature
(B) acidic nature
(C) H- bonding
(D) Low volatility

Qix. One molal aqueous solution of which salt has the lowest freezing point .....

(A) NaCl
(B) KBr
(C) $CaCl_2$
(D) $Al_2 (SO_4)_3$

Qx. A property of liquids that decreases with increase in intermolecular forces, is ....

(A) Viscosity
(B) Boiling point
(C) surface tension
(D) Vapor pressure

Qxi. When two moles of SO_2 react with one mole of O_2 to form 2 moles of SO_3, the limiting reactant will be .....

(A) SO_2
(B) O_2
(C) SO_3
(D) none of these

Qxii. A substance with regular and orderly arranged particles is .....

(A) glass
(B) ice
(C) water
(D) steam

Qxiii. A solution is acidic when .....

(A) pH > 7
(B) pOH < 7
(C) $[OH^-] > 1 \times 10^{-7}$
(D) $[H^+] > 1 \times 10^{-7}$

Qxiv. Total mole fraction (x) of a solution is always .....

(A) x = 0
(B) x = 1
(C) x >1
(D) x <1

Qxv. The sum of exponents of concentration terms in the rate equation for a reaction is .....

(A) rate
(B) rate constant
(C) molecularity
(D) order

Qxvi. The oxidation state of phosphorus in P_4 molecule is .....

(A) zero
(B) +4
(C) -4
(D) +5

Qxvii. Moles of NH_3 produced when one mole of each nitrogen and hydrogen react together

(A) 1 mol
(B) 2 mol
(C) 2/3 mol
(D) 1/3 mol

Qxviii. A quantum number that explains the fine structure of an atom is .....

(A) n
(B) l
(C) m
(D) ms

Section "B"

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

QQ.2. Attempt any TEN parts. All parts carry equal marks.

i Why 4s orbital is filled first with electrons than 3d orbital?

ii What is meant by absolute zero?.

iii What is meant by buffer action?.

iv Why NaCl is non-conductor in solid state but becomes good conductor in molten state?.

v Define amphoteric substance?. How water is amphoteric in nature?.

vi Define common ion effect?. Why it happens?.

vii How temperature affects the rate of a chemical reaction?.

viii Calculate the Molarity of 500ml aqueous solution containing 20g NaOH.

ix How Born Haber cycle helps to calculate the lattice energy of a binary ionic solid?.

x Calculate the value of 51g NH_3 gas at STP?.

xi What are bonding and anti bonding molecular orbitals?.

xii Define oxidizing and reducing agents. Give suitable examples.

xiii What is collision theory?. Write its important postulates.

Section "C"

Maximum Marks: 27

Note: Note: Attempt any THREE questions. All questions carry equal marks.

QQ.3.

a [5] What are X-rays?. How X-rays are produced?

b [4] Explain 1st and 2nd order reactions with suitable examples.

QQ.4.

a [5] State and explain, law of mass action.

b [4] Calculate the molar mass of an unknown gas which diffuses two times less rapidly than hydrogen gas.

QQ.5.

a [5] Discuss London forces in detail.

b [4] Why oxygen molecule is Paramagnetic in nature?.

QQ.6.

a [4] Explain Isomorphism and Polymorphism.

b [5] What is lowering of vapor pressure. How lowering of vapor pressure helps to determine the molecular mass of solute in dilute solution. .